



Genetic Prediction

Polygenic scores, embryo selection, and where this might be going

How far can you get with a linear model and a few million genotypes?

A thought experiment

Two newborns are accidentally switched at birth.



Child A

Born to a family of professors —
raised by a family with no interest in books or
ideas.



Child B

Born to the anti-intellectual family —
raised by the professors.

Who ends up smarter at age 30?

Heritability is larger than many people believe

Nature has run this experiment, at scale.



Twin studies

Identical twins raised apart end up nearly as similar in adult IQ as twins raised together.

0.6–0.8

heritability of adult cognitive ability — and it rises with age (“Wilson effect”).



Adoption studies

By adulthood, adoptees' IQs correlate substantially with their biological parents — and only weakly with their adoptive parents.

~ a few points

the typical IQ boost from adoption into a more advantaged home — real, but not a transformation.

Caveat: these estimates apply within the range of ordinary, non-deprived environments. Severe deprivation is a different story.

Surprising linearity

High heritability isn't just high — it's surprisingly additive.



Breeding works

Selective breeding works reliably in cattle, crops, and dogs — which tells you the genotype→phenotype map is, in aggregate, close to additive.



Additive variance dominates

For height, cognitive ability, and common-disease liability, dominance and gene–gene interaction effects turn out to be small in practice.



A plain linear model is near-optimal

You can build a near-optimal genetic predictor with a linear model over SNPs. Fancier architectures have produced only marginal gains so far.



One practical detail

Regress out the first ~10–20 genetic principal components first — they capture ancestry / population structure, which is confounded with environment and culture.

Counterintuitive — if the genome is code



Genome as source code

Imagine predicting what a program does by summing an independent score for each character in the source file. Absurd.



And biology is nonlinear

At the molecular level everything is regulatory networks, feedback loops, and thresholds — about as nonlinear as it gets.

So why on earth do linear predictors work?

A continuous function, viewed locally



Locally linear regime

The trait is some complicated continuous function of thousands of small inputs. Each variant nudges the input slightly, and the population occupies a small neighborhood of genotype space — so the first-order Taylor expansion captures most of the story.



Central-limit flavor

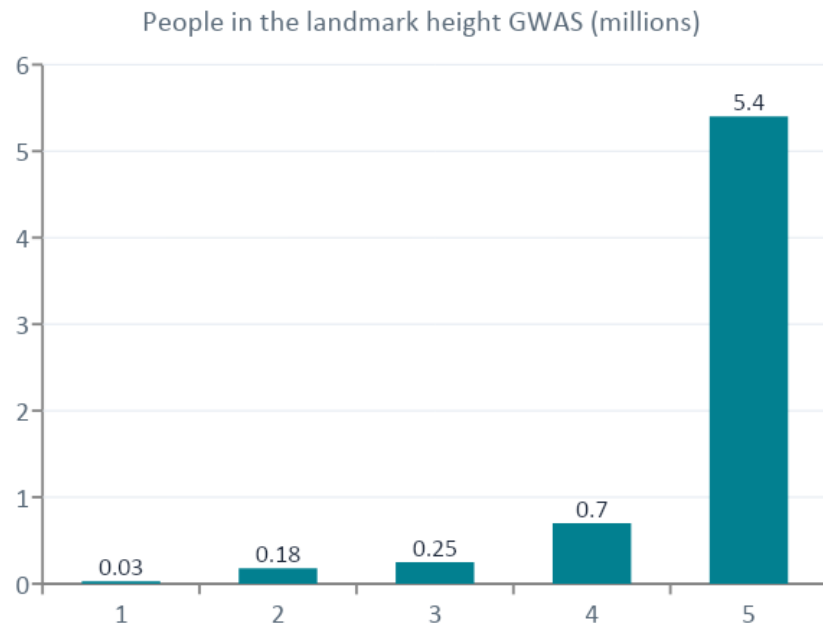
Thousands of small, roughly independent contributions sum to something well-described by an additive model — even when individual pathways are nonlinear.



For the ML crowd: model souping

Interpolating between trained networks in weight space “shouldn’t” work if you think of weights as code — but it does, because the relevant region is flat and connected. Genomes within a population are like checkpoints fine-tuned from a common ancestor.

What training data is there?



- UK Biobank: ~500k genotyped, deeply phenotyped — the workhorse.
- 23andMe: >10M genotyped customers (subset consented for research).
- Million Veteran Program ~1M · FinnGen ~500k · All of Us ~1M target.
- Biobank Japan, China Kadoorie, Estonian Biobank, and more.

Why the chart stops at 2022: for common variants, height is now saturated — the frontier has moved to whole-genome sequencing (UK Biobank's ~500k genomes, All of Us), richer phenotypes, and new ancestries.

But how can *you* get training data?



Individual-level genomes are locked up

Access committees, data-use agreements, air-gapped enclaves — and for good reason. You can't just download a million genomes.



Open-source summary statistics (“sumstats”)

Per-SNP effect sizes, standard errors, and allele frequencies from published GWAS — freely downloadable for hundreds of traits.



They aggregate across biobanks

Cohorts that could never share raw genotypes can share sumstats and meta-analyze. Privacy risk is much lower (though not literally zero).



This simplifies things tremendously

A grad student with a laptop, public sumstats, and one LD reference panel can build a state-of-the-art predictor.

How do you actually learn the weights?



With individual-level data

Just run a sparse penalized regression — lasso or elastic net over ~a million SNPs vs. the phenotype. (Roughly what's done inside UK Biobank.)



With sumstats only

Marginal effects double-count signal because nearby SNPs are correlated (LD). Un-smear them with lassosum (a sumstats-only lasso), or Bayesian MCMC methods — LDpred2, PRS-CS, SBayesRC (adds functional annotations; roughly state of the art).



The output is just a weight vector

$\text{score} = \sum (\text{allele dosage} \times \text{weight})$. Scoring a new genome takes milliseconds.



Or skip the homework: the PGS Catalog

pgscatalog.org hosts thousands of published score files, free to download, plus tooling to apply them — for many traits you never fit anything yourself. (An open pipeline built on these is coming up in a few slides.)

Applications



For an already-existing individual: honestly underwhelming

Pharmacogenetics is mostly a handful of single-gene variants. Consumer reports: “slightly elevated diabetes risk; also your eyes are probably brown.” Modest clinical value in risk stratification (earlier screening for the top few % of polygenic risk).



For livestock: transformative

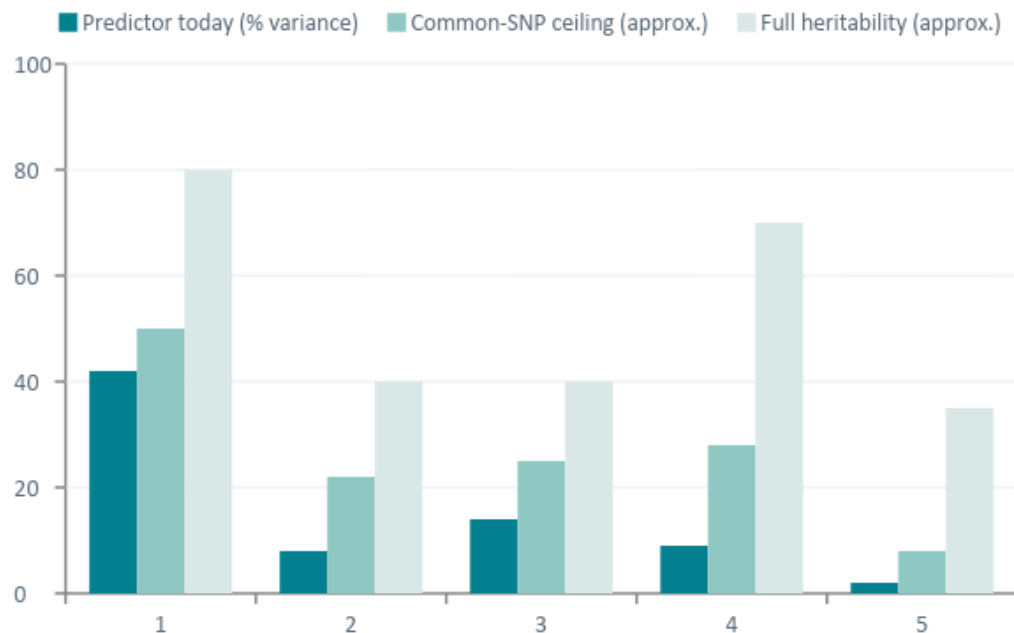
Genomic selection roughly doubled the rate of genetic gain in dairy cattle. The bull stud business is now essentially a polygenic-score business.



The interesting human application: prospective parents

You can't change your own genome — but you can influence which genome your child gets. That's the rest of this talk.

How good are current predictors?



- Height: ~40–45% — the common-SNP signal is now essentially saturated (2022 GWAS, $n = 5.4M$). The gap to ~80% is rare variants & family effects.
- Disease scores: top-percentile individuals at ~3–5× average risk for CAD, T2D, breast cancer.
- Cognition / education: meaningful, far from either ceiling.
- Subjective well-being: ~1–2% — a rounding error so far.

Caveat — ancestry portability: accuracy drops with genetic distance from the (mostly European) training data — roughly half the predictive power in East/South Asian and admixed-American samples, and often only ~20–40% of it in African-ancestry samples. Mostly an artifact of who's in the training data, not of biology.

More data: how much would it help?



Gains are forecastable

PGS accuracy as a function of N follows a known curve (heritability + effective number of causal variants). Height is nearing its ceiling; cognitive traits are far from theirs — tens of millions of well-phenotyped samples could plausibly take cognitive prediction from ~10% to 30–40%+ of variance.



Where new data comes from

Biobanks keep growing and federating (sumstats make that easy); consumer genomics sits on huge untapped cohorts; whole-genome sequencing recovers rare-variant signal; better phenotyping (real cognitive tests, longitudinal records) matters as much as more genomes.



Diversify ancestry

Closes the portability gap and improves fine-mapping for everyone.

What can a couple doing IVF expect today?



Few embryos

A typical cycle yields ~3–5 viable, euploid embryos (maybe 10 in a good case). Only ~55% of transfers reach live birth, so the effective batch is smaller still: $n_{\text{eff}} \approx 0.55 \cdot n$.



Embryos are siblings

They share ~half their segregating variants, so within-batch genetic variance is half the population's (SD ~70%).



You select on the predictor, not the trait

Gains get discounted by predictor accuracy.

$\approx 2.5\text{--}3 \text{ cm}$

expected height gain, best of 10 embryos, today's predictors

$\approx 2.5\text{--}3 \text{ IQ pts}$

expected cognitive-ability gain
(best of 5: $\approx 1.5\text{--}2$)



Non-European-ancestry couples: multiply the gains by ≈ 0.75 (East/South Asian, Hispanic/Latino) or ≈ 0.5 (African ancestry) — gains scale with $\sqrt{\text{relative predictor accuracy}}$, and today's predictors travel poorly.

Back-of-envelope: $\text{gain} \approx c(n_{\text{eff}}) \cdot \sqrt{(r^2/2)} \cdot \text{SD}$, with $c(5) \approx 1.2$ and $c(10) \approx 1.5$. A failed transfer drops you to the runner-up, so 10 embryos behave like ~ 5.5 and the figures above are post-attrition (Karavani et al. 2019).

Shameless plug: an open pipeline for all of this

github.com/dawndrain/genetic_prediction — ~2k lines of Python, MIT-licensed.



Embryo selection, end to end

Phased parental genomes + sparse PGT-A biopsy reads ($\sim 0.05\times$) $\rightarrow$ joint HMM recovers each embryo's genotypes $\rightarrow$ score ~ 20 traits $\rightarrow$ rank.



Phasing is the hard part

At a realistic $\sim 1\%$ switch-error rate, a naive HMM recovers only $r^2 \approx 0.4$ of the score; a joint multi-embryo HMM gets ≈ 0.9 , one genotyped relative ≈ 1.0 .



Side quest

Also scores 23andMe files / VCFs against curated PGS Catalog scores, with imputation, ancestry adjustment, and openSNP validation.

+0.35 QALY

expected gain, best of 5 embryos, today's predictors (Monte Carlo over ~ 20 traits)

$\approx \frac{1}{4}$

of the additive-heritability ceiling — the gap is mostly rare variants GWAS can't see yet

Only $\sim 55\%$ of transfers reach live birth, so 5 embryos behave like $n_{\text{eff}} \approx 2.75$. Demo embryo report and the predictor-scaling analysis are linked from the README.

Strong selection (speculative from here on)

Everything from here on is blocked on wet-lab advances, not on statistics.



More embryos: in vitro oogenesis

Mature large numbers of eggs from reprogrammed cells (works in mice, not yet humans) and select from ~1,000 embryos instead of ~5: best-of-1,000 $\approx$ 3.2 SD vs. best-of-5 $\approx$ 1.2 SD on the predictor — roughly 3× the gain. Attrition barely dents this: you keep ~95% of the gain at 1,000 embryos vs. ~62% at 5. Funding momentum comes from infertility, not selection.



Sperm selection: sequence the meiotic “twins”

You can't sequence a sperm without destroying it — but each meiosis yields four products that stay physically connected. Sequence the discarded siblings (plus dad's phased genome) and the survivor's genome follows by complementarity; for eggs, the polar-body version already exists. Sperm are abundant, so selection is strong — but each sperm is only half the child's genome: best-of-thousands $\approx$ +1.5–2 SD on the score, and it stacks with embryo selection.

Strong selection, continued

Still wet-lab-blocked — but the ceiling is much higher.



Per-chromosome selection

Choose which copy of each chromosome gets passed on: ~ 23 independent lotteries instead of one shuffle, so gains scale like $\sqrt{23} \approx 5\times$. No credible lab path today — a thought experiment about where the ceiling is.



Iterated selection

Differentiate embryonic cells back into gametes and run select $\rightarrow$ cross $\rightarrow$ select: ~ 10 generations of selection compressed into a couple of years. Effects stop being “a few points” — and the ethical questions stop being hypothetical.

This is the regime where the effects — and the societal questions — stop being small.

Things worth saying out loud

- A polygenic score is a probabilistic nudge on a distribution, not a destiny — environment still matters, and within-family prediction is noisier than between-family.
- The ancestry gap means early benefits (and early markets) accrue unevenly.
- Strong selection raises real questions — access, which traits get optimized, who decides — that the field will not get to skip.
- The statistics are basically solved and the data keeps growing. The binding constraints are wet-lab biology and societal choices. Which moves first?

Open embryo-selection pipeline, validation, and QALY calculator: github.com/dawndrain/genetic_prediction